

VYTEGRA, Russian Federation

WMO: 228370

Lat: 61.0168N

Lon: 36.4556E

Elev: 56

StdP: 100.65

Time Zone: 3.00 (E03)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.2	(c) -26.2	(d) -32.7	(e) 0.2	(f) -30.1	(g) -28.7	(h) 0.3	(i) -26.0	(j) 9.0	(k) -4.5	(l) 7.7	(m) -3.4	(n) 0.4	(o) 10	(p) 0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.9	(c) 27.4	(d) 19.3	(e) 24.9	(f) 17.6	(g) 23.0	(h) 16.7	(i) 19.8	(j) 25.4	(k) 18.8	(l) 23.7	(m) 17.6	(n) 21.7	(o) 2.7	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	12.9	22.6	16.8	12.0	21.0	15.8	11.3	19.8	57.0	25.3	53.4	23.8	49.8	21.5	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.5	5.8	-33.3	30.6	3.3	2.2	-35.7	32.2	-37.7	33.5	-39.5	34.7	-41.9	36.3
			-33.4	21.6	3.3	1.4	-35.8	22.6	-37.7	23.4	-39.6	24.2	-42.0	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-8.9	-8.6	-3.1	3.2	8.0	14.6	16.6	14.3	9.2	3.6	-3.8	-6.8		
	DBStd	10.56	7.55	8.58	4.95	4.98	4.99	4.39	3.47	3.17	4.26	4.72	6.18	7.20		
	HDD10.0	3038	586	520	405	211	101	7	1	4	66	203	414	521		
	HDD18.3	5545	844	753	663	454	323	129	75	127	276	457	664	779		
	CDD10.0	581	0	0	0	7	37	146	205	139	41	4	0	0		
	CDD18.3	44	0	0	0	0	1	18	21	3	1	0	0	0		
	CDH23.3	389	0	0	0	1	17	187	157	23	4	0	0	0		
	CDH26.7	92	0	0	0	0	3	53	35	0	0	0	0	0		
(14) Wind	WSAvg	2.5	2.7	2.7	2.7	2.5	2.5	2.3	2.2	2.1	2.4	2.6	2.7	2.7		
(15) Precipitation	PrecAvg	677	42	31	32	35	47	67	79	88	70	75	61	52		
	PrecMax	944	65	76	81	95	101	147	226	224	144	170	119	113		
	PrecMin	492	10	6	2	4	3	8	19	14	16	0	14	15		
	PrecStd	90	14	15	16	19	20	30	44	41	28	35	24	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.8	5.1	9.6	21.5	25.3	31.1	29.9	25.6	23.7	16.8	8.1	3.9		
		MCWB	1.6	3.3	5.7	12.2	16.2	21.1	19.8	17.6	17.6	13.7	6.9	3.1		
	2%	DB	2.0	3.5	5.9	17.7	21.8	28.1	27.4	23.6	20.1	13.5	6.0	3.0		
		MCWB	1.0	2.4	2.7	10.7	14.0	19.9	19.2	16.7	15.8	10.9	5.1	2.2		
	5%	DB	1.3	2.2	4.3	14.3	19.2	25.4	24.7	22.0	17.2	11.2	4.8	2.2		
		MCWB	0.4	1.1	1.9	8.6	12.8	18.1	17.7	16.6	13.9	9.2	3.9	1.3		
	10%	DB	0.6	1.2	3.1	11.2	16.7	22.9	22.9	20.4	15.0	9.9	3.2	1.3		
		MCWB	-0.2	0.3	1.3	6.9	11.6	16.5	17.1	15.6	12.7	8.5	2.6	0.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	3.6	5.7	12.8	17.7	21.7	21.1	19.2	18.3	13.9	6.8	3.3		
		MCDB	2.3	4.7	8.8	20.5	24.1	29.0	27.4	23.3	22.2	17.0	7.6	3.6		
	2%	WB	1.2	2.5	3.5	11.0	15.4	20.3	19.8	18.0	16.1	11.1	5.3	2.3		
		MCDB	1.8	3.4	5.0	17.0	19.7	26.7	25.2	21.6	19.4	13.1	6.0	2.8		
	5%	WB	0.6	1.2	2.4	9.1	13.4	18.9	18.9	17.1	14.4	9.6	4.0	1.4		
		MCDB	1.2	2.0	3.8	13.4	18.6	24.4	23.8	20.5	16.8	10.9	4.8	2.1		
	10%	WB	-0.1	0.4	1.4	7.2	11.7	17.4	17.6	16.3	12.9	8.4	2.6	0.6		
		MCDB	0.6	1.3	2.8	10.9	16.6	21.5	21.8	19.7	15.1	9.8	3.3	1.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.1	7.7	8.4	8.8	9.5	8.9	7.6	6.5	4.9	4.7	5.4		
		MCDBR	4.4	4.7	7.7	13.3	13.1	12.9	12.3	10.9	9.3	6.8	4.0	3.3		
	5% WB	MCWBR	4.4	4.1	5.4	7.7	7.2	6.5	5.5	5.9	5.5	4.7	3.6	3.2		
		MCWBR	4.2	4.7	5.3	12.5	12.1	11.7	10.4	9.2	8.4	6.1	3.4	3.6		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.303	0.327	0.368	0.358	0.354	0.391	0.377	0.344	0.325	0.279	0.228		
		taud	2.036	2.074	2.130	2.259	2.433	2.469	2.385	2.403	2.478	2.383	2.132	1.984		
	Edn at Noon		449	649	777	821	866	875	827	808	776	648	421	313		
		Edh at Noon	42	77	102	110	100	99	105	95	74	58	38	27		
(44) All-Sky Solar Radiation	RadAvg		0.26	0.87	2.22	3.78	5.02	5.32	5.12	3.89	2.22	0.89	0.25	0.13		
	RadStd		0.04	0.22	0.30	0.42	0.33	0.34	0.62	0.29	0.22	0.12	0.04	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.57	N/A	N/A	N/A	N/A	N/A	-219	-232	N/A	N/A

Nomenclature: See separate page